



**School Of Engineering
Mechanical Engineering Department**

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I HEREBY RECOMMEND THAT THE CAPSTONE PREPARED UNDER MY
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ENTITLED

Variable Rate Adaptive Sprayer System

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Variable Rate Adaptive Sprayer System

Capstone submitted in partial fulfillment of the requirements

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Dedicated To,

My Family, Friends, Mentors and loved ones who have been in my corner through every
step of the way.

Abstract

The current labour and resource shortage is driving the agriculture industry towards adopting more humane, sustainable and efficient means of production. Robotics and automation is one of the solutions to address those challenges. Precision spraying, in particular, is a technique that relies on a precise targeted metered delivery of a desired fluid. In this paper we have designed a sprayer system as a module for our in-house modular rover that can administer a fluid in a controlled manner. It offers a discrete left and right independent spray control, each with the range of 120 - 400 L/h. Tests conducted in the field show the performance within 0.1 - 4.8 % of the commanded value. Furthermore, we layout the framework for a high level sprayer-rover interface controller to intelligently adapt and command the desired flow rates

Keywords: Variable Rate Sprayer, Precision Spraying, Precision Agriculture

Contents

Acknowledgments	iii
Abstract	v
Contents	v
List of Figures	vi
List of Tables	vii
1 Introduction	1
1.1 Ethical Implications and Considerations	2
2 System Overview	4
2.1 Mechanical Subsystem	4
2.2 Electrical Subsystem	5
2.3 Software Subsystem	6
2.4 Sprayer-Rover Integration	6
3 Control and Validation	8
3.1 Control System Design	8
3.1.1 Mapping Function	10
3.2 Testing and Results	12
3.2.1 Left Bank Flow: 131 L/h & Right Bank Flow: 131 L/h	12
3.2.2 Left Bank Flow: 360 L/h & Right Bank Flow: 210 L/h	13
3.2.3 Left Bank Flow: 210 L/h & Right Bank Flow: 360 L/h	14
3.2.4 Continuous Variable Flow Rate Test Run	15
4 Conclusion & Future Work	17
4.1 Conclusion	17
4.2 Future Work	17
A Appendix	20
A.1 Code Listings	20
A.2 Conference Paper	28

List of Figures

2.1	AgBot Rover and Sprayer System in the Field	5
2.2	Sprayer system overview.	5
2.3	Sprayer System Plumbing and Instrumentation Diagram	6
2.4	Sprayer and the rover system integration.	7
3.1	Control Block diagram representation of Sprayer System.	9
3.2	Sprayer system left-bank open loop response at different valve positions. . .	10
3.3	Sprayer system open-loop nozzle flow rate left and right bank time varying response to a ramp (pump power) input at a constant valve position of 80%. .	11
3.4	Step-Input response to desired flow rate inputs of 131 l/h for both left and right bank.	13
3.5	Step-Input response to desired flow rate inputs of 360 l/h for the left bank and 210 l/h for the right bank.	14
3.6	Step-Input response to desired flow rate inputs of 210 l/h for the left bank and 360 l/h for the right bank.	14
3.7	Multi-zone time response.	15
A.1		29
A.2		30
A.3		31
A.4		32
A.5		33
A.6		34
A.7		35

List of Tables

3.1 Zone-level flow rates (L/h) 16

Chapter 1

Introduction

Agricultural industries across the globe play vital roles when it comes to the nourishment and survival of humankind. Feeding an ever-growing global population places ever-increasing demands on an already stressed industry: one affected by labor shortages, climate change, human labor rights concerns, stricter food safety regulations, and a desire for more sustainable farming practices (reducing food waste) [1]. The continually expanding application of robotics and automation technologies within the agricultural sector has created a viable solution to the aforementioned challenges; the subsequent practice is referred to as precision agriculture. Robotics and automation technologies integration within agriculture can lead to more efficient, cost-effective, sustainable, and humane farming practices [2].

The application of pesticides, herbicides, fungicides, and/or other “-cides” is a significant process in agriculture pertaining to the management of pests, diseases, and weeds, allowing for increased yields. Efficient application of the aforementioned agrochemicals increases yield, but can cause environmental risks in the event of over application or double coverage of a given region. Additionally, over application can pose health risks to humans as well [3]. Over application incurs greater costs on the farmers as well. Precision spraying is one domain within precision agriculture which is adopting automation technologies to improve efficacy and minimize waste via various precision spraying techniques and methodologies.

Variable rate automated sprayer systems consist of two major processes: automated

vision-based weed detection and automated targeted-spraying; it is an area of current research and development efforts [4], [5], [6], [7]. The aforementioned techniques have been implemented in large-scale production fields, displaying the viability of the technology [8]. Commercial systems, such as the One Smart Spray from Bosch [9] employ real-time weed detection via high-resolution cameras to identify weeds; it then applies herbicide in the locations needed, reducing costs for the growers while also reducing the amount of herbicide needed.

Different vehicle types exist within the realm of precision spraying: unmanned aerial vehicles (UAV) [10], ground-level fluid lines, and self-propelled ground vehicles. The scope of the presented work is to create a modular precision sprayer system which is able to be integrated to an existing rover, developed by Santa Clara University's Robotics Systems Laboratory; the rover is referred to as the Agbot [11]. The motivation behind the creation of the sprayer system is to elevate the functional capabilities of the Agbot and similar robotic/rover platforms which serve as a support system for various precision agricultural operations. [12].

1.1 Ethical Implications and Considerations

Robots have shown their utility in agricultural field operations [13]. Capable of operating in environments that are deemed harsh and difficult for humans, as well as being able to meet the demands of strenuous labor requirements and challenging work patterns, robots and robotic devices are capable of supplanting human laborers. The pace of the transition will likely be a gradual process, creating a transition phase in which humans and robots will co-exist in the field operations [14]. Another considerable challenge the agriculture industry faces is a significant labor shortage; the consequences of which includes crops going unharvested, leading to food waste, and higher labor costs [15]. Utilizing robotic and automation technology can supplement and provide the existing labor and its capabilities, but with added efficiency at a more cost-effective rate. Despite all the benefits robotics and automation is poised to provide, we acknowledge that there are some ethical considerations. Like most transitions and adoptions of new technologies, there are positive and

negative impacts. Supplanting human labors with robotics, especially in precision spraying application, reduces humans' exposure to spraying chemicals, decreasing the health risks of the farmers. Additionally, robotics and automation are able to perform repetitive, heavy-labor tasks, alleviating human laborers from the chore. The tradeoff is that human laborers will be displaced.

Chapter 2

System Overview

The system overview is split in four different subsection. The first three subsections provides an overview of the mechanical, electrical and software architecture of the sprayer module followed by a description on the specifics of how the sprayer module is integrated within the rover system in mechanical, electrical and software sense. Figure [A.7](#)

2.1 Mechanical Subsystem

The mechanical sub-system is comprised of two sub-assemblies: the frame structure and the plumbing. The frame structure serves as the backbone of the sprayer system. It is assembled from aluminum T-slotted rails. Virtually all of the subsystems are mounted onto the frame structure. The plumbing sub system includes (1) Fluid Reservoir, (2) positive-displacement pumps, silicone vacuum seal tubing, and (4) modular Nozzle Units (NUs) as shown in Figure 1.

The NUs are comprised of (3) key components: a servo valve, a flow sensor, and a nozzle. NU 1, NU 2, and the Left Pump constitute the left bank of the Sprayer System. NU 3, NU 4, and the Right Pump make up the right bank. Each bank operates independently of the other: they only share the same reservoir tank. This is shown in plumbing schematic.

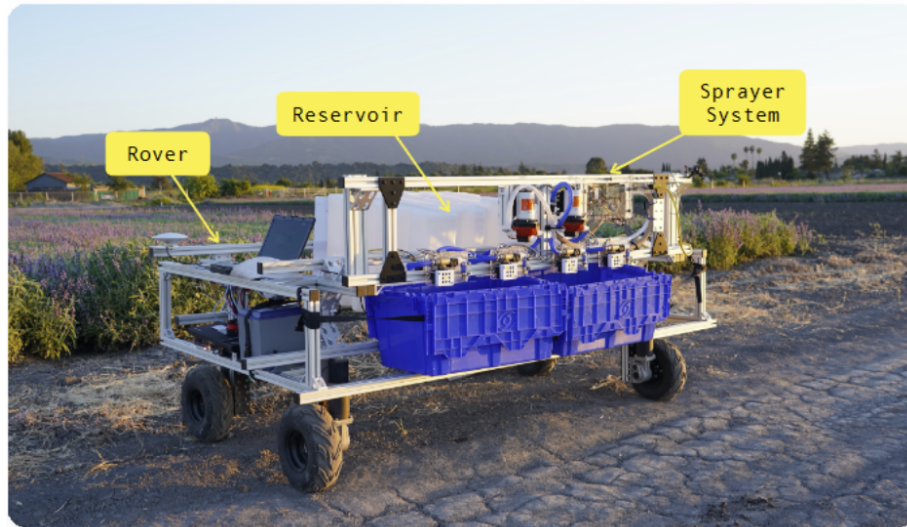


Figure 2.1 – AgBot Rover and Sprayer System in the Field

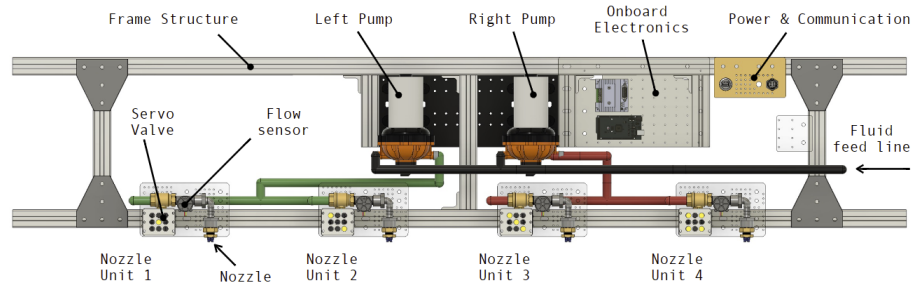


Figure 2.2 – Sprayer system overview.

2.2 Electrical Subsystem

The key electrical subsystem comprises of the pump, servo controlled valve, and flow rate sensor. The two pumps are controlled by a (RoboteQ) motor controller. An Arduino Mega 2560 is the micro-controller of choice which controls the respective pump power, servo-valve, and receives the instantaneous flow rate information from all four nozzle units.

An onboard computer (Jetson Nano) is dedicated to allow the information exchange between the sprayer system and the rover system. The entire sprayer system is designed to operate at 12 VDC which is provided by the rover.

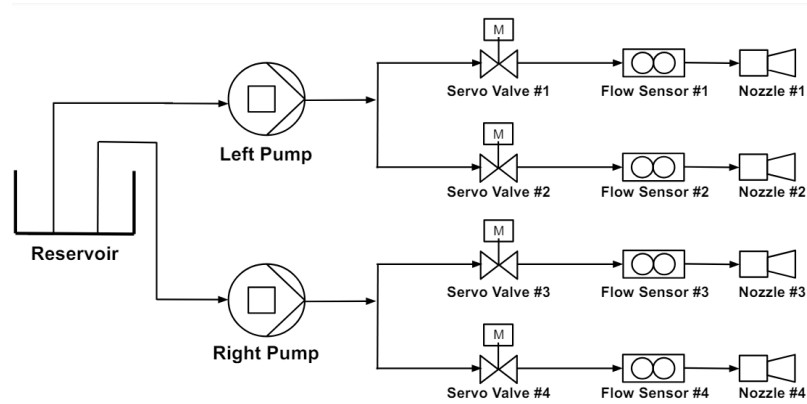


Figure 2.3 – Sprayer System Plumbing and Instrumentation Diagram

2.3 Software Subsystem

Three functional systems exist within the software subsystem: data acquisition, data display, and the controller. Real-time flow rate data and pump power is recorded for each flow sensor and each pump. It is stored in ROS 1 bags via an additional Arduino Mega 2560. The aforementioned real-time data is displayed in a serial monitor within the Arduino IDE.

An error-based Proportional Integral (PI) controller with a feed-forward term is implemented for the sprayer system; desired flow rate data and measured flow rate data is used to calculate the error.

2.4 Sprayer-Rover Integration

In a *bus* and *payload* analogy, the sprayer system is the payload which relies on the rover (the *bus*) to receive the necessary support needed for the mission. Both, the rover and the sprayer system are constructed using a standard size T-slotted aluminium bar. This allows for a convenient mechanical integration using off-the-shelf brackets. The sprayer receives the electrical power from the rover. The rover system uses the Robot Operating System (*ROS*) middleware as the main control framework [11]. Figure 2.4 shows the overall sprayer-rover integrated system. The sprayer controller has access to the position sensors, speed, vision system and other onboard data to compute desired flow rate information.

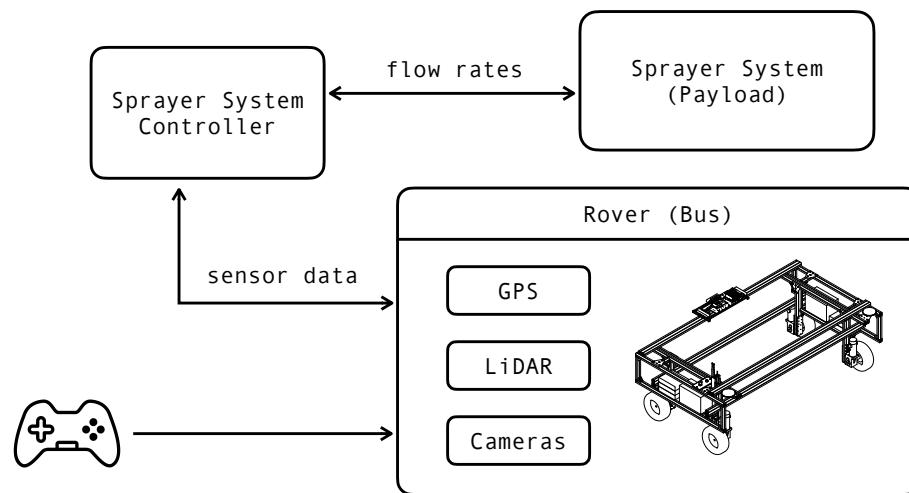


Figure 2.4 – Sprayer and the rover system integration.

Chapter 3

Control and Validation

While Proportional-Integral-Derivative (PID) control methodology is a very established and robust control scheme, we opted for a Proportional-Integral (PI) Control with a mapping function. The mapping function serves as a pseudo feed-forward term which incorporates information regarding the open-loop behavior of the system.

3.1 Control System Design

The control system design is a Proportional-Integral (PI) controller with the addition of a mapping function. The controller scheme inputs are desired flow rates for the left and right bank, $\bar{q}_l$ and $\bar{q}_r$, respectively. Separate and unique mapping functions M_l and M_r are used to convert the desired flow rates into a pump powers for each bank. The desired flow rate signals are then summed with the measured flow rates, q_l and q_r creating resultant error signals, e_l and e_r for the two banks. The error signal is then fed into a Proportional-Integral controller. The output of the left and right bank controllers yield a pump power value, p_l and p_r . The pump power values are then summed at each summing junction and fed into the their respective pumps. The pumps then produce a flow rate, which is fed through the servo valves, flow sensor, and nozzle; this ultimately outputs the measured flow rate.

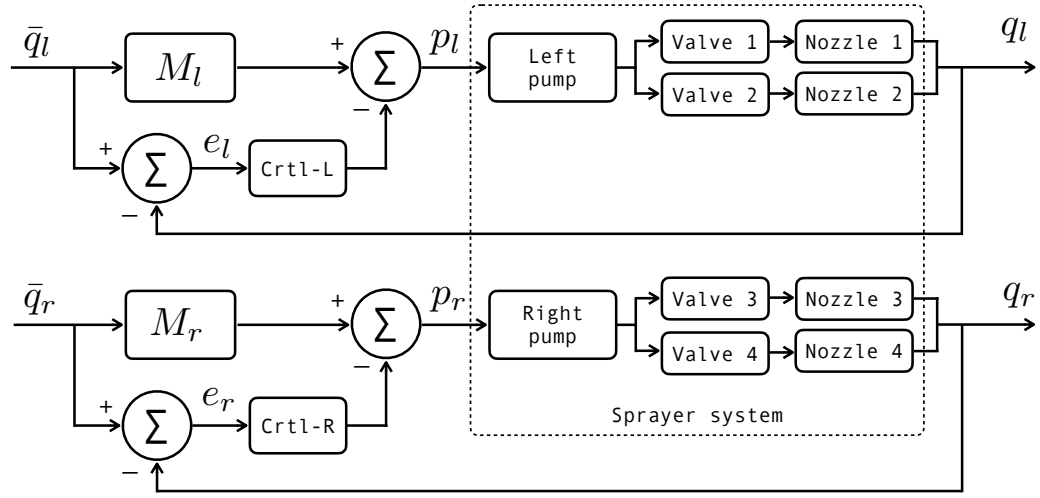


Figure 3.1 – Control Block diagram representation of Sprayer System.

$$M_l : \hat{p}_l \mapsto q_l$$

$$M_r : \hat{p}_r \mapsto q_r$$

Given the desired left and right flow rates ($\bar{q}_l$ and $\bar{q}_r$) the respective errors are

$$e_l = \bar{q}_l - q_l, \text{ and}$$

$$e_r = \bar{q}_r - q_r.$$

The respective pump powers are

$$u_l = p_l = M_l^{-1} \bar{q}_l + (k_{p_l} e_l + k_{i_l} \int e_l), \text{ and}$$

$$u_r = p_r = M_r^{-1} \bar{q}_r + (k_{p_r} e_r + k_{i_r} \int e_r).$$

where K_{p_l} and K_{p_r} are the proportional terms of the left bank controller and the right bank controller, respectively. K_{i_l} and K_{i_r} represent the integral terms of the left and right bank, respectively.

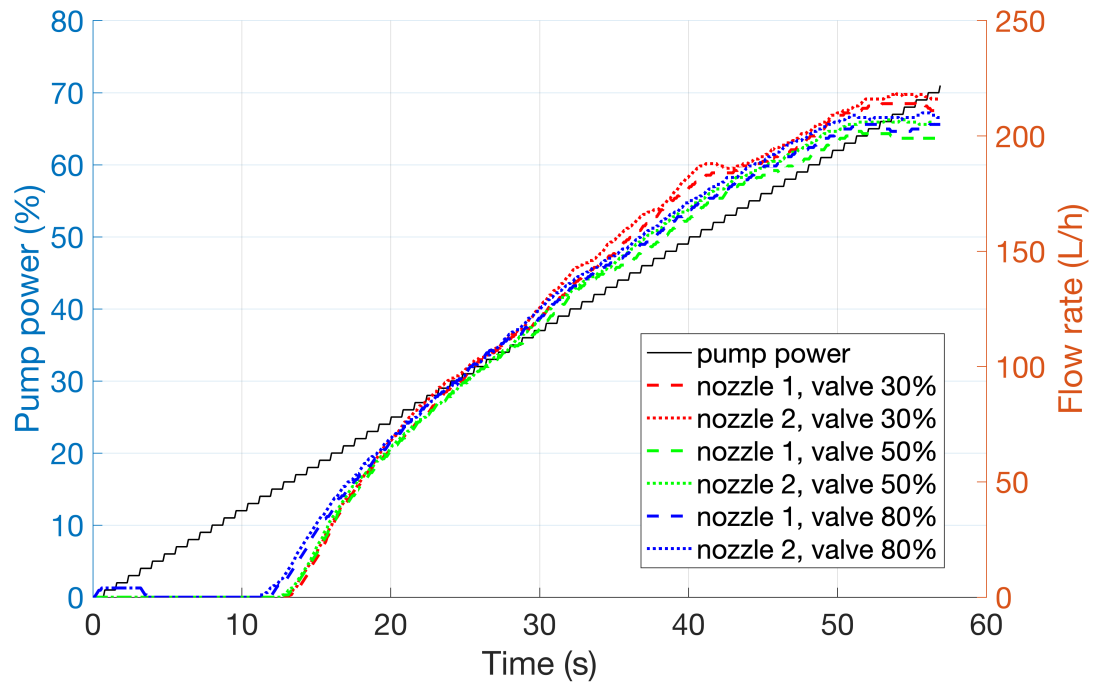


Figure 3.2 – Sprayer system left-bank open loop response at different valve positions.

3.1.1 Mapping Function

The relationship between valve positions, pump power, and flow rate was investigated. An open-loop system test was ran for a ramp-input of pump powers ranging from 0% to 70%. Three of these open-loop trials were ran, each at a constant valve position. In Figure 3.2, the open-loop response of the Left Bank can be seen. It is discovered that for a given pump power, the valve position has virtually no effect on the flow rate of the system – barring the valve is completely closed. Pump power is the only variable that affected the flow rate.

Utilizing the open-loop characteristics of the system, distinct mapping functions are created for each bank, correlating pump power to a specific flow rate. The plot was generated by using a ramp-input at valve positions of 80% open. Figure 3.3 displays the open loop behavior of each bank. It can be seen that despite being the same pump, the open-loop characteristics of each of the banks differs slightly; for the same pump power, the Right Bank is able to dispense for fluid than the Left Bank. This could be due to differing lengths of tubing needing to be used or that the Right Bank pump was new compared to the Left Bank pump.

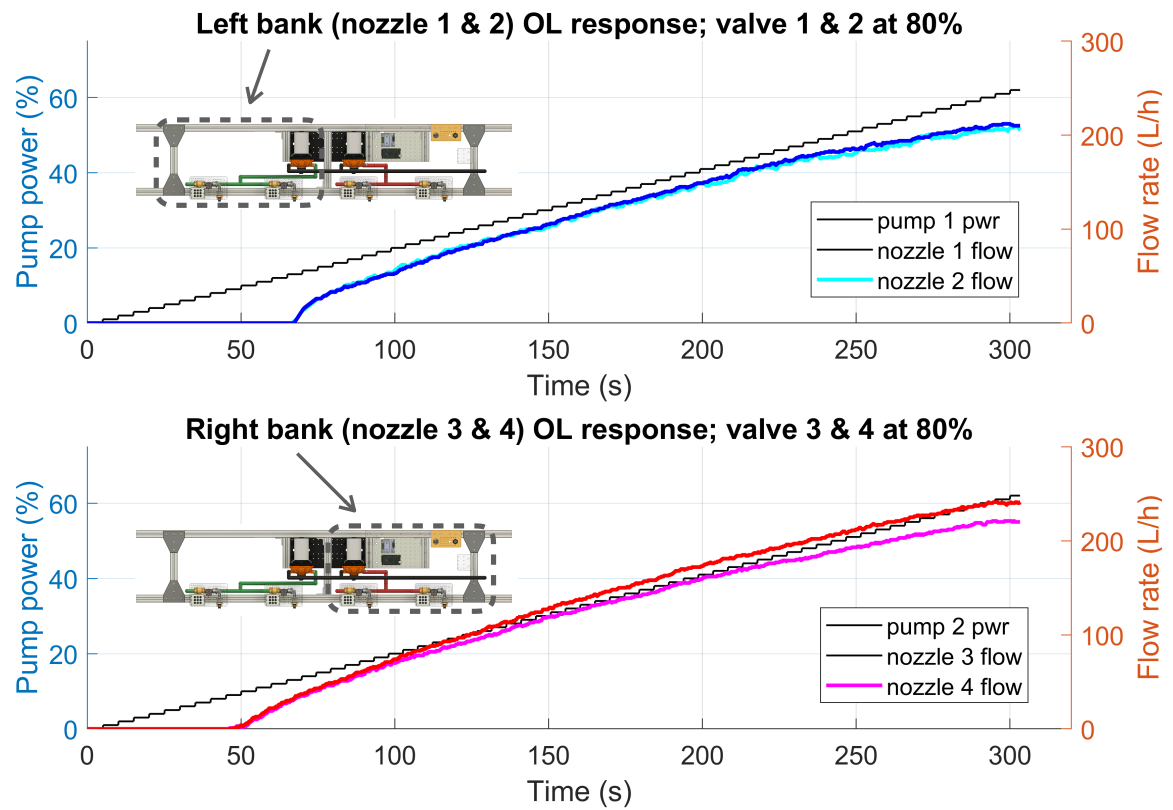


Figure 3.3 – Sprayer system open-loop nozzle flow rate left and right bank time varying response to a ramp (pump power) input at a constant valve position of 80%.

3.2 Testing and Results

The sprayer is a closed loop system, delivering a desired flow rate using an error-based controller. The performance of the system is highly dependent on the accuracy of each flow sensor. System performance verification was performed concurrently with system performance testing.

System performance testing was solely based on the collecting flow sensor data given an array of desired inputs. The verification portion involved collecting the dispensed water via collection bins and noting the total time elapsed during the trial. Average flow rates for each bank were then calculated by dividing the total volume of water collected in the bins by the total time of the trial. Note that this method of verification is prone to human error in measurement and reaction time, resolution error in the analog measuring tools, evaporation due to wind, etc.

The testing occurred on a nearby farm in an environment that would likely resemble the rover and sprayer system's natural area of operation: one with external factors such as wind, dust, and uneven, rocky terrain. The testing occurred while the rover was in motion at a speed within the range of $0.2m/s$ to $0.4m/s$. In order to determine the performance of the system, time-response plots are captured for arbitrary step-inputs.

3.2.1 Left Bank Flow: 131 L/h & Right Bank Flow: 131 L/h

In Fig. 3.4, the desired flow is set to 131 L/h for both banks. It takes the left and right bank approximately 8.5 seconds to settle given a 15% error band. The reason behind the relatively large error band is because the system is not able to fully settle below a 15% error due to limitations in the sensors' resolution and computation.

The recorded flow rate data appears to take on discrete values. In the steady state region of Fig. 3.4, the recorded data takes on one of the following values: 113 L/h, 122 L/h, 131 L/h, a 140 L/h. This indicates that the resolution of the sensor is roughly 9L/h per bit. Given this finding, performing Root Mean Squared (RMS) calculations on the entirety of data provides greater insight over the steady state behavior.

The RMS value for the left bank in Fig. 3.4 is 125.3 L/h, and the RMS value for the right

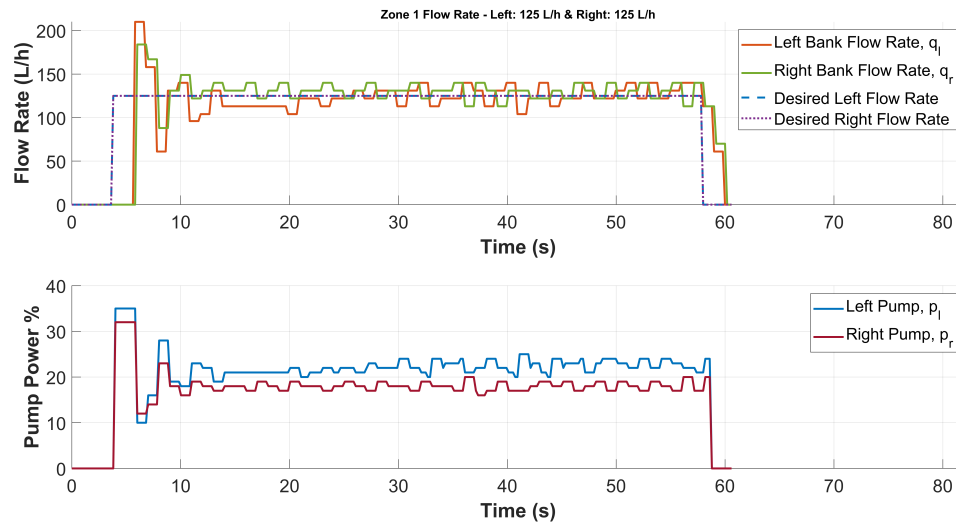


Figure 3.4 – Step-Input response to desired flow rate inputs of 131 l/h for both left and right bank.

bank is 130.8 L/h. The RMS value is compared against the desired flow rate of 131L/h for both banks; it equates to an error of 4.8% and 0.1% for the left and right bank, respectively.

From the verification test, the calculated flow rates are determined to be 133.4 L/h and 131.7 L/h for the left and right banks, respectively. This amounts to an error of 1.8% and 0.5% between the desired flow rates and averaged flow rates of the system.

3.2.2 Left Bank Flow: 360 L/h & Right Bank Flow: 210 L/h

In Fig.3.5, the settling time for the system is shorter, and the error band is smaller; for a 10% error band, the settling time is approximately 2.5 seconds for both banks. The RMS values are 353.4L/h and 211.0L/h for the left and right bank, respectively. It equates to errors of 1.8% and -0.4% .

For the verification test performed in Fig. 3.5, the desired flow rates are 360L/h and 210L/h, respectively. The average flow rates are calculated to be 356.0L/h and 211.8L/h for the left and right bank respectively: yielding absolute errors of 1.1% and 0.8%

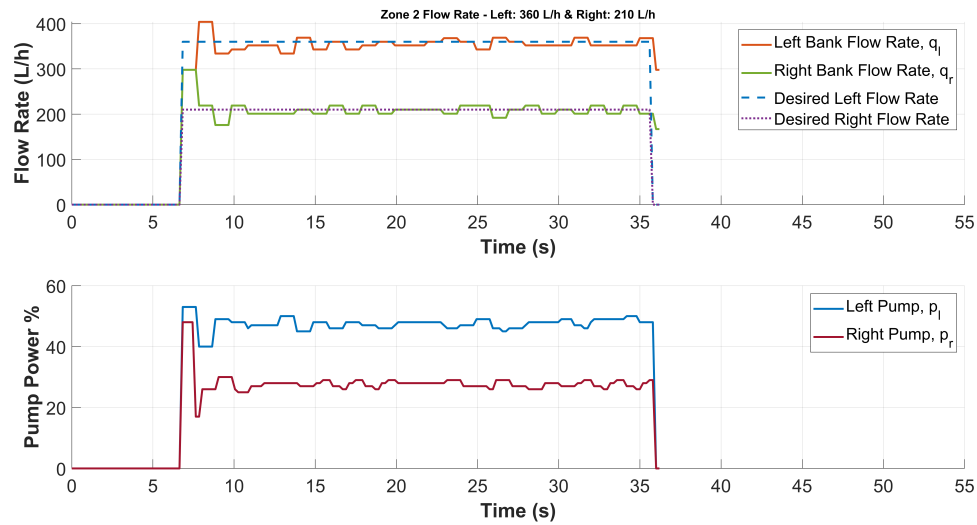


Figure 3.5 – Step-Input response to desired flow rate inputs of 360 l/h for the left bank and 210 l/h for the right bank.

3.2.3 Left Bank Flow: 210 L/h & Right Bank Flow: 360 L/h

In Fig.3.6, a similar result can be seen when the desired flow rates for the left and right banks are swapped with the other. The 10% settling time is also approximately 2.5 seconds for both banks. The RMS values are 208.0L/h and 353.4L/h for the left and right bank, respectively. This yields errors of 0.9% for the left bank and 1.8% for right bank.

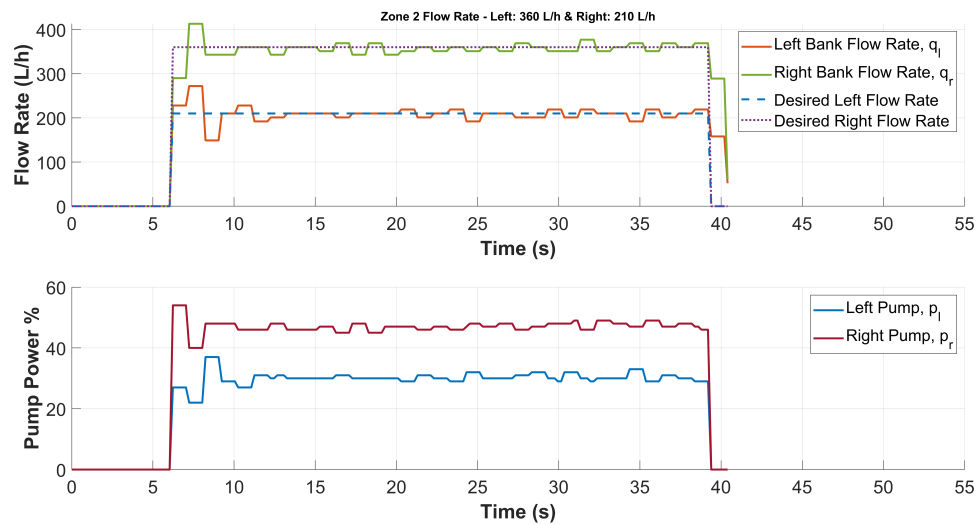


Figure 3.6 – Step-Input response to desired flow rate inputs of 210 l/h for the left bank and 360 l/h for the right bank.

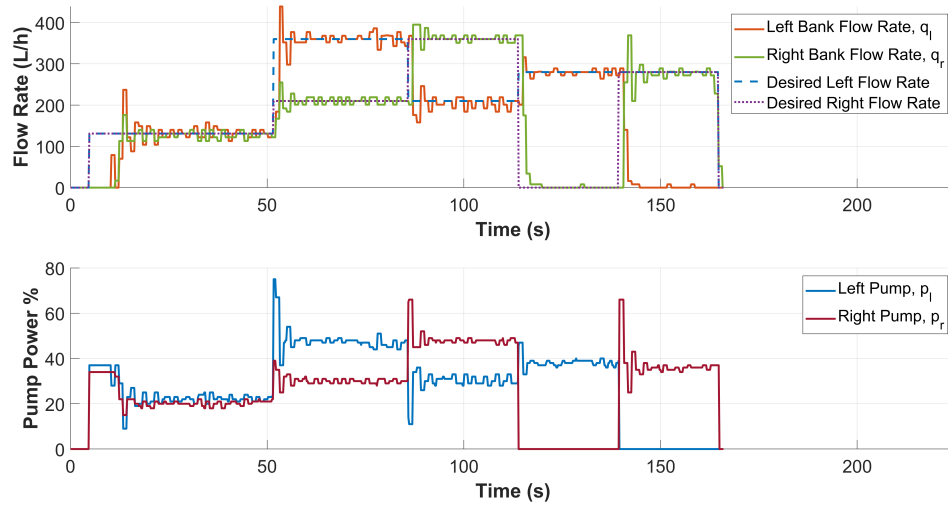


Figure 3.7 – Multi-zone time response.

In Fig. 3.6 the desired flow rates for the left and right bank are 210L/h and 360L/h , respectively. The average flow rates for the left and right bank were 217.5L/h and 352.3L/h , amounting to errors of 3.6% and 2.1%.

3.2.4 Continuous Variable Flow Rate Test Run

Figure 3.7 shows a performance test for a multi-zone trial. Five arbitrary zones were created, each with different desired flow rates. The multi-zone trial was performed while the rover was moving at speeds between 0.2m/s and 0.4m/s . The zones were advanced via button toggling through the main rover controller; this displays the potential ability to control the sprayer through an outer, rover control loop which could include GPS data, LIDAR, or vision data as shown in Fig. 2.4

The performance of the multi-zone run is consistent with what was seen in the step-input responses. The desired flow rate, sensor reading, and actual measured volume/calculated flow rates are tabulated in 3.1. In terms of system performance, the settling time between zone transitions is between 2 to 3 seconds; it remains consistent with the individual step inputs. One notable exception can be seen in the plot during the initial power up; the reason behind the larger than expected settling time of the first zone is due to the pumps needing time to prime the system. Left at rest for a sufficient amount of time, the water

Table 3.1 – Zone-level flow rates (L/h)

Zone	Left-bank flow			Right-bank flow		
	Desired Flow	Sensor	Actual Flow	Desired Flow	Sensor	Actual Flow
1	131	125.3	133.4	131.0	130.8	131.7
2	360.0	353.4	356.0	210.0	211.0	211.8
3	210.0	208.0	217.5	360.0	353.4	352.3

drains downward and out of pumps, thus causing a larger initial settling time.

The results from the system performance tests indicate that the system is able to administer the desired flow rate within a 4.8% RMS error margin. (i.e. RMS flow sensor reading compared to the desired flow rate). The system performance verification shows that the administered flow rate is accurate to within a 3.6% error margin (i.e. the physical amount of water collected over a finite time compared to the desired flow rate)

Chapter 4

Conclusion & Future Work

4.1 Conclusion

In this paper We have designed and constructed a modular sprayer attachment for the existing Agbot Rover which provides the necessary infrastructure for the mission. The sprayer system's closed-loop proportional-integral controller with a mapping function has been shown to be capable of administering the desired flow rate within an error margin between 0.1% and 4.8%. Furthermore it has displayed the ability to toggle between various different flow rates within a 2 – 3 second settling time between each transition. Lastly, a framework has been created between the sprayer and rover which will allow for the exchange of information between the two. The main purpose of the framework is to develop prescription-based controller.

4.2 Future Work

The next logical step of the Sprayer System and Agbot Rover evolution would be to incorporate GPS data from the Agbot Rover to control the desired flow rate based on the location of the Rover in field. Further future work could be focused on incorporating the Rover's velocity in the sprayer system controller in order to provide a flow rate metric of volume sprayed per area; this is the industry standard with regards to spraying operations within the agriculture industry.

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Appendix A

Appendix

This Capstone Paper is predominately based on a conference paper written for an ASME Conference. The Capstone paper goes into further detail regarding the controller used for the sprayer system.

A.1 Code Listings

The listings package allows you to include code listings or other formatted text with some parsing to make them more readable than simply calling `\input{}` on the code file.

Listing A.1 – ARDUINO Sprayer System Controller Code label

```
// Hardware - Arduino Mega 2560

int u_p1 = 0;
int u_p2 = 0;

float pump_pwr[2] = {0.0, 0.0};
float valve_pos[4] = {0.0, 0.0, 0.0, 0.0};
float flow_rate_uf[4] = {0.0, 0.0, 0.0, 0.0};
float flow_rate[4] = {0.0, 0.0, 0.0, 0.0};
float des_flow_rate[4] = {0.0, 0.0, 0.0, 0.0};

const int window_size = 400; // define the window size.
int flow_s0_ar[window_size];
int flow_s1_ar[window_size];
int flow_s2_ar[window_size];
```



```
int flow_s3_ar[window_size];

// pins specific to DF robot 4ch driver shield, DRI0039
// https://wiki.dfrobot.com/Quad_Motor_Driver_Shield_for_Arduino_SKU_DRI0039
const int E1 = 3; // valve1 pwm
const int E2 = 11; // valve2 pwm
const int E3 = 5; // valve3 pwm
const int E4 = 6; // valve4 pwm
const int M1 = 4; // valve1 dir
const int M2 = 12; // valve2 dir
const int M3 = 8; // valve3 dir
const int M4 = 7; // valve4 dir


volatile int flow_frequency18, flow_frequency19, flow_frequency20, flow_frequency21 ;
// Measures flow sensor pulses
unsigned int l_hour18, l_hour19, l_hour20, l_hour21; // Calculated litres/hour
const byte flowsensor18 = 18; // Flow Sensor Input 1
const byte flowsensor19 = 19; // Flow Sensor Input 2
const byte flowsensor20 = 20; // Flow Sensor Input 3
const byte flowsensor21 = 21; // Flow Sensor Input 4
unsigned long currentTime;
unsigned long cloopTime;
unsigned long cloopTime2;


void flow18 () // Interrupt function
{
    flow_frequency18++;
}
void flow19 () // Interrupt function
{
    flow_frequency19++;
}
void flow20 () // Interrupt function
{
    flow_frequency20++;
}
void flow21 () // Interrupt function
{
    flow_frequency21++;
}


void set_valve(int i = 0, int x = 0)
{

```

```
digitalWrite (M1,LOW);
digitalWrite (M2,HIGH);
digitalWrite (M3,LOW);
digitalWrite (M4,HIGH);

if (x < 0) x = 0;
else if (x >= 100) x = 100;
int v = map(x, 0, 100, 0, 255);

if (i == 1) analogWrite(E1, v);
if (i == 2) analogWrite(E2, v);
if (i == 3) analogWrite(E3, v);
if (i == 4) analogWrite(E4, v);
}

void ctrl_pump(int ch = 0, int val = 0)
{
    int min_val_pos = 20;
    if (ch == 1)
    {
        if (valve_pos[0] < min_val_pos && valve_pos[1] < min_val_pos)
            val = 0;
    }
    if (ch == 2)
    {
        if (valve_pos[2] < min_val_pos && valve_pos[3] < min_val_pos)
            val = 0;
    }

    int max_per = 60;
    if (val < 0) val = 0;
    else if (val > max_per) val = max_per;
    int v = map(val, 0, 100, 0, 1000);

    if (ch==1)
    {
        Serial3.print("!G_1_");
        Serial3.print(String(v));
        Serial3.print("_\r");
    }

    if (ch==2)
    {
        Serial3.print("!G_2_");
```

```
        Serial3.print(String(v));
        Serial3.print("_\r");
    }
}

void debug_disp()
{
    Serial.print("des"); Serial.print("\t");
    Serial.print(des_flow_rate[0], 1); Serial.print("\t");
    Serial.print(des_flow_rate[1], 1); Serial.print("\t");
    Serial.print(des_flow_rate[2], 1); Serial.print("\t");
    Serial.print(des_flow_rate[3], 1); Serial.print("\t");
    Serial.println();

    Serial.print("flo"); Serial.print("\t");
    Serial.print(flow_rate[0], 1); Serial.print("\t");
    Serial.print(flow_rate[1], 1); Serial.print("\t");
    Serial.print(flow_rate[2], 1); Serial.print("\t");
    Serial.print(flow_rate[3], 1); Serial.print("\t");
    Serial.println();

    Serial.print("vlv"); Serial.print("\t");
    Serial.print(valve_pos[0],0); Serial.print("\t");
    Serial.print(valve_pos[1],0); Serial.print("\t");
    Serial.print(valve_pos[2],0); Serial.print("\t");
    Serial.print(valve_pos[3],0); Serial.print("\t");
    Serial.println();

    Serial.print("pmp"); Serial.print("\t");
    Serial.print(pump_pwr[0], 0); Serial.print("\t");
    Serial.print(pump_pwr[1], 0); Serial.print("\t");
    Serial.println();
    Serial.println(u_p1);
}

float maf_s0(float x)
{
    int p = window_size;
    flow_s0_ar[p] = x;
    float a = 0;
    for(int i = 1; i <= p; i++)
        a += flow_s0_ar[i];
    a = a/p;
    for(int i = 2; i <= p; i++)
        flow_s0_ar[i-1] = flow_s0_ar[i];
}
```

```
    return a;
}

float maf_s1(float x)
{
    int p = window_size;
    flow_s1_ar[p] = x;
    float a = 0;
    for(int i = 1; i <= p; i++)
        a += flow_s1_ar[i];
    a = a/p;
    for(int i = 2; i <= p; i++)
        flow_s1_ar[i-1] = flow_s1_ar[i];
    return a;
}

float maf_s2(float x)
{
    int p = window_size;
    flow_s2_ar[p] = x;
    float a = 0;
    for(int i = 1; i <= p; i++)
        a += flow_s2_ar[i];
    a = a/p;
    for(int i = 2; i <= p; i++)
        flow_s2_ar[i-1] = flow_s2_ar[i];
    return a;
}

float maf_s3(float x)
{
    int p = window_size;
    flow_s3_ar[p] = x;
    float a = 0;
    for(int i = 1; i <= p; i++)
        a += flow_s3_ar[i];
    a = a/p;
    for(int i = 2; i <= p; i++)
        flow_s3_ar[i-1] = flow_s3_ar[i];
    return a;
}

void get_flow_rates()
{
    currentTime = millis();
    // Every second, calculate and print litres/hour
    if (currentTime >= (cloopTime + 1000))
```

```

{
  cloopTime = currentTime; // Updates cloopTime
  // Pulse frequency (Hz) = 23Q, Q is flow rate in L/min.

  l_hour18 = ((flow_frequency18 * 60 / 7.5) * 1.1);
  // (Pulse frequency x 60 min) / 23Q = flowrate in L/hour
  flow_frequency18 = 0; // Reset Counter
  flow_rate_uf[0] = l_hour18;

  l_hour19 = ((flow_frequency19 * 60 / 7.5) * 1.1);
  // (Pulse frequency x 60 min) / 23Q = flowrate in L/hour
  flow_frequency19 = 0; // Reset Counter
  flow_rate_uf[1] = l_hour19;

  l_hour20 = ((flow_frequency20 * 60 / 7.5) * 1.1);
  // (Pulse frequency x 60 min) / 23Q = flowrate in L/hour
  flow_frequency20 = 0; // Reset Counter
  flow_rate_uf[2] = l_hour20;

  l_hour21 = ((flow_frequency21 * 60 / 7.5) * 1.1);
  // (Pulse frequency x 60 min) / 23Q = flowrate in L/hour
  flow_frequency21 = 0; // Reset Counter
  flow_rate_uf[3] = l_hour21;

  debug_disp();
}

// moving average filter
if (currentTime >= (clloopTime2 + 1000))
{
  flow_rate[0] = maf_s0(flow_rate_uf[0]);
  flow_rate[1] = maf_s1(flow_rate_uf[1]);
  flow_rate[2] = maf_s2(flow_rate_uf[2]);
  flow_rate[3] = maf_s3(flow_rate_uf[3]);
}

}

void setup()
{
  pinMode(flowsensor18, INPUT_PULLUP); // Optional Internal Pull-Up
  pinMode(flowsensor19, INPUT_PULLUP); // Optional Internal Pull-Up

```

```

pinMode(flowsensor20 , INPUT_PULLUP);  // Optional Internal Pull-Up
pinMode(flowsensor21 , INPUT_PULLUP);  // Optional Internal Pull-Up

attachInterrupt(digitalPinToInterrupt(flowsensor18), flow18,  RISING); // Setup Interrupt
attachInterrupt(digitalPinToInterrupt(flowsensor19), flow19,  RISING); // Setup Interrupt
attachInterrupt(digitalPinToInterrupt(flowsensor20), flow20,  RISING); // Setup Interrupt
attachInterrupt(digitalPinToInterrupt(flowsensor21), flow21,  RISING); // Setup Interrupt
currentTime = millis();
cloopTime = currentTime;

Serial.begin(9600);
Serial3.begin(115260);  // roboteq controller

// declare valve pins
for(int i=3;i<9;i++)
    pinMode(i,OUTPUT);
for(int i=11;i<13;i++)
    pinMode(i,OUTPUT);

int x = 50; // set valves to 0
set_valve(1,x);
set_valve(2,x);
set_valve(3,x);
set_valve(4,x);

}

void loop ()
{
    get_flow_rates();
    ctrl_pump(1, pump_pwr[0]); ctrl_pump(2, pump_pwr[1]);
    set_valve(1, valve_pos[0]); set_valve(2, valve_pos[1]);
    set_valve(3, valve_pos[2]); set_valve(4, valve_pos[3]);

    if (Serial.available())
    {
        /*
            int x = Serial.parseInt();
            int y = Serial.parseInt();
            int z = Serial.parseInt();

```

```
    if (x == 1)
    {
        valve_pos[0] = y;
    }
    if (x == 2)
    {
        valve_pos[1] = y;
    }
    if (x == 3)
    {
        valve_pos[2] = y;
    }
    if (x == 4)
    {
        valve_pos[3] = y;
    }
    pump_pwr[0] = z;
    pump_pwr[1] = z;
    */

    des_flow_rate[0]= Serial.parseInt();
    des_flow_rate[1]= Serial.parseInt();
    des_flow_rate[2]= Serial.parseInt();
    des_flow_rate[3]= Serial.parseInt();
}

valve_pos[0] = 0;
valve_pos[1] = 0;
valve_pos[2] = 0;
valve_pos[3] = 0;

int e1 = des_flow_rate[0] - flow_rate_uf[0];
int e2 = des_flow_rate[2] - flow_rate_uf[2];
float k1 = 0.3;
float k2 = 0.3;
u_p1 = k1 * e1;
u_p2 = k2 * e2;

if (u_p1 > 60) u_p1= 60;
if (u_p1 < 0) u_p1= 0;

if (u_p2 > 60) u_p2= 60;
if (u_p2 < 0) u_p2= 0;

pump_pwr[0] = u_p1;
```

```
pump_pwr[1] = u_p2;
```

```
}
```

A.2 Conference Paper

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VARIABLE RATE ADAPTIVE SPRAYER SYSTEM

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ABSTRACT

The current labour and resource shortage is driving the agriculture industry towards adopting more humane, sustainable and efficient means of production. Robotics and automation is one of the solutions to address those challenges. Precision spraying, in particular, is a technique that relies on a precise targeted metered delivery of a desired fluid. In this paper we have designed a sprayer system as a module for our in-house modular rover that can administer a fluid in a controlled manner. It offers a discrete left and right independent spray control, each with the range of 120 - 400 L/h. Tests conducted in the field shows the performance within 0.1 - 4.8 % of the commanded value. Furthermore, we layout the framework for a high level sprayer-rover interface controller to intelligently adapt and command the desired flow rates.

Keywords: Precision spraying, Variable flow rate, Prescription fluid administration

1. INTRODUCTION

The advent of robotics and automation in the field of agriculture helps in the emergence of novel techniques which were previously impossible due to technical limitations and high cost. Among others, precision spraying is one such ag domain that is adopting technology to improve efficiency and emphasize sustainable practices. Current research in the area of automated sprayer systems includes automated vision-based weed detection and targeted spraying [1–3]. These techniques have also been implemented in a large production field showcasing the viability of precision spraying [4]. Other notable work in the field of variable rate precision sprayer include [5–8]. One Smart Spray [9] from Bosch is a precision sprayer system that leverages high-resolution cameras to detect weeds in real time and spray herbicide only where needed, reducing the use and cost of inputs for growers.

In terms of type of spray vehicle, there are three broad categories namely, network of fluid lines laid in ground, aerial vehicle [10], and self-propelled vehicle system. The work presented here is under the category of self-propelled system which is designed as a modular subsystem to an existing rover system, called Agbot [11] developed in the Robotic Systems Lab. The motivation in designing this sprayer system was to elevate the functionality of the pre-existing Agbot which is designed in-house [11] at Santa Clara University. This is the very first exterior tool created to integrate with the Agbot, or other similar robotic platform [12], which acts as a support system to back the sprayer.

Multiple pre-existing systems were examined and are broken down into performance factors in Table 1.

1.1 Ethical Implications

Agriculture is one of the areas where robots have started to show their potential in various field operations [18, 19]. Labor-intensive work profiles and challenging work conditions may be addressed through robotic devices and potential co-exist during the transition phase [20], thereby allowing laborers to be re-allocated to higher skilled work. Furthermore, the agriculture industry itself faces a dramatic shortage of labor, leading to high labor costs and unharvested crops [21]. Robotic and automation technology can also be used to expand the use of precision agriculture practices in order to lead to more efficient, cost-effective and sustainable farming practices [22].

Whilst this sprayer design was created with the greater good in mind, we acknowledge that it poses some ethical issues. Like most things, automating the agriculture spraying process has both positive and negative impacts. Some positive aspects include decreasing health risk to farmers when spraying chemicals, decreasing the amount of heavy labor farmers have to perform, and specified input control on how much is spraying each section, which minimizes waste. Some negative impacts include possible displacement of labor, reduction of farmer's autonomy, and a potential increase of damaging impacts of agricultural runoff by increasing production.

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Figure A.1

TABLE 1: Analysis of pre-existing sprayer systems

Legend	Performance	Leading feature	Limitations	Cite
See & Spray Pre-mium (John Deere)	Hit rate of 12 mph	Cameras & ML ID weeds for precision targeting	No dual-product solution system or split tank	[13, 14]
One Smart Spray (Bosch)	95% application accuracy	Weed targeting Green on green & green on brown performance LED lighting to spray night & day	Limited to corn, soy, cotton, sunflower, & canola	[9]
Bilberry (Trimble)	Reduces herbicide usage by 90% Up to 25 km/h rover speed	Weed targeting Green on green & green on brown performance	Green on green & limited to & certain weeds: & Grass, Broadleaf, and Blue Lupin in Lupins, etc	[15, 16]
Greeneye Technology	Reduces herbicide usage by 90% Up to 25 km/h rover speed	Weed targeting, Classifies weeds by species Green on green & green on brown performance		[17]

1.2 Contributions

1. Modular sprayer system for a rover,
2. Test and validate the spray performance, and
3. Create a foundation for a prescription based spray controller.

2. DESIGN PHASE

During the early developmental phase we considered a sprayer system with eight nozzles spread equidistantly to cover a furrow width of approximately 1.5 meters. A single pump was chosen as the main actuator to move the fluid. Each nozzle receives the fluid from the common pump outlet with a solenoid-plunger valve in series respectively. The valves can be commanded to a fully open or fully close position to control the flow rate by modifying the width of each pulse. During the testing phase we discovered that the highly discrete nature due to the low valve fidelity results in a spray pattern where there are large gaps of no-spray zone. Additionally, the pump of choice is a positive displacement pump which means the pump power directly relates to the flow rate. This results in a highly coupled system where flow rate of each nozzles affect one another.

On the other hand, a centrifugal pump generates a constant pressure thought and as the nozzles start to dispense the volume flow is quickly and inherently replenished as a result of the constant pressure. Therefore, the individual nozzles operate in a relative independent and less coupled. This is similar to how a common rail direct injection system works in a conventional internal combustion engines.

There are several merits and demerits of using a positive displacement pump over a centrifugal pump. We have chosen to use a positive displacement pump due to its safe low voltage operation, compact size and lack of priming. For the purpose of proof of concept demonstration, based on our previous work we have decided to have a sprayer system consisting of four nozzles driven by two independent pumps. In the following sections we describe the sprayer system in more details.

3. SPRAYER SYSTEM OVERVIEW

The system overview is split in four different subsection. The first three subsections provides an overview of the mechanical, electrical and software architecture of the sprayer module followed by a description on the specifics of how the sprayer module is integrated within the rover system in mechanical, electrical and software sense.

3.1 Mechanical Sub-system

The mechanical sub-system is comprised of two systems: the frame-structure and the plumbing. The frame structure serves as the backbone of the sprayer system. It is assembled from aluminum T-slotted rails. Virtually all of the subsystems are mounted onto the frame structure. The plumbing sub-system includes (1) Fluid Reservoir, (2) positive-displacement pumps, silicone vacuum seal tubing, and (4) modular Nozzle Units (NUs) as shown in Figure 1. The NUs are comprised of (3) key components: a servo valve, a flow sensor, and a nozzle.

NU 1, NU 2, and the Left Pump constitute the left bank of the Sprayer System. NU 3, NU 4, and the Right Pump make up the right bank. Each bank operates independently of the other: they only share the same reservoir tank. This is shown in plumbing schematic.

3.2 Electrical Sub-system

The key electrical subsystem comprises of the pump, servo controlled valve, and flow rate sensor. The two pumps are con-

Figure A.2

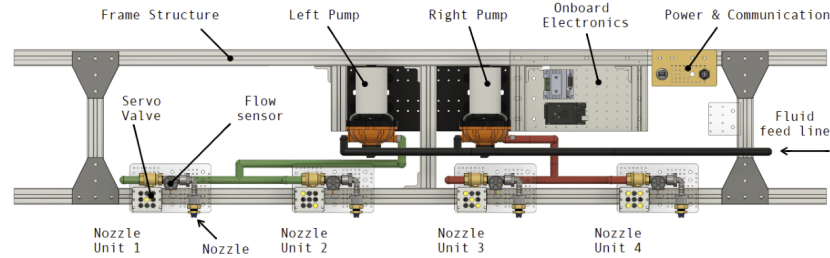


FIGURE 1: Sprayer system overview.

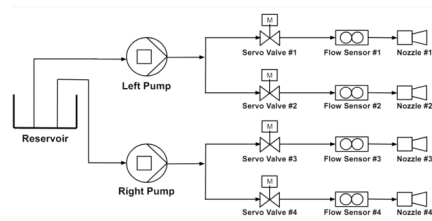


FIGURE 2: Sprayer System Plumbing and Instrumentation Diagram

trolled by a (RoboteQ) motor controller. An Arduino Mega 2560 is the microcontroller of choice which controls the respective pump power, servo-valve, and receives the instantaneous flow rate information from all four nozzle units. An onboard computer (Jetson Nano) is dedicated to allow the information exchange between the sprayer system and the rover system. The entire sprayer system is designed to operate at 12 VDC which is provided by the rover.

3.3 Software Sub-system

Three functional systems exist within the software subsystem: data acquisition, data display, and the controller. Real-time flow rate data and pump power is recorded for each flow sensor and each pump. It is stored in ROS 1 bags via an additional Arduino Mega 2560. The aforementioned real-time data is displayed in a serial monitor within the Arduino IDE. An error-based Proportional Integral (PI) controller with a feed-forward term is implemented for the sprayer system; desired flow rate data and measured flow rate data is used to calculate the error.

3.4 Sprayer-Rover Integration

In a *bus* and *payload* analogy, the sprayer system is the payload which relies on the rover (the *bus*) to receive the necessary support needed for the mission. Both, the rover and the sprayer system are constructed using a standard size T-slotted aluminium bar. This allows for a convenient mechanical integration using off-the-shelf brackets. The sprayer receives the electrical power from the rover. The rover system uses the Robot Operating System

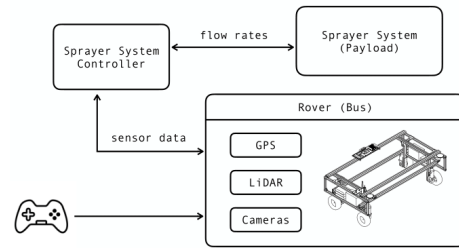


FIGURE 3: Sprayer and the rover system integration.



FIGURE 4: Sprayer system with rover.

(ROS) middleware as the main control framework [11]. Figure 3 shows the overall sprayer-rover integrated system. The sprayer controller has access to the position sensors, speed, vision system and other onboard data to compute desired flow rate information.

4. CONTROL AND VALIDATION

4.1 Control System Design

The control system design scheme inputs are desired flow rates for the left and right bank, $\bar{q}_l$ and $\bar{q}_r$, respectively. Separate and unique mapping functions are used to convert the desired flow

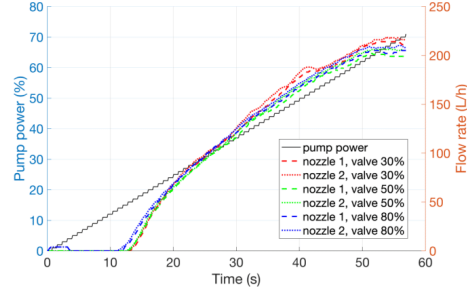


FIGURE 5: Sprayer system left-bank open loop response at different valve positions.

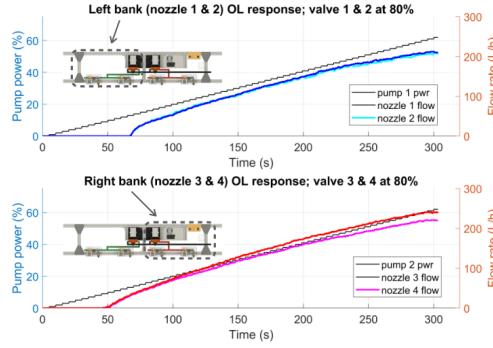


FIGURE 6: Sprayer system open-loop nozzle flow rate left and right bank time varying response to a ramp (pump power) input at a constant valve position of 80%.

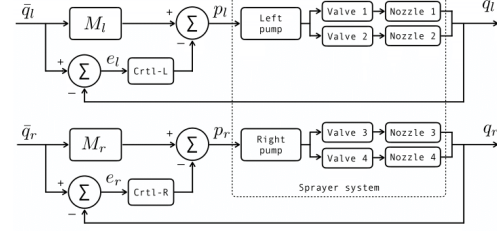


FIGURE 7: Block diagram representation of sprayer control system.

rates into a pump powers for each bank. The desired flow rate signals are then summed with the measured flow rates, creating resultant error signals for the two banks. The error signal is then fed into a Proportional-Integral controller. The output of the left and right bank controllers yield a pump power value. The pump power values are then summed at each summing junction and fed into their respective pumps. The pumps then produce a flow rate, which is fed through the servo valves, flow sensor, and nozzle; this ultimately outputs the measured flow rate.

$$M_l : \hat{p}_l \mapsto q_l$$

$$M_r : \hat{p}_r \mapsto q_r$$

Given the desired left and right flow rates ($\bar{q}_l$ and $\bar{q}_r$) the respective errors are

$$e_l = \bar{q}_l - q_l, \text{ and}$$

$$e_r = \bar{q}_r - q_r.$$

The respective pump powers are

$$u_l = p_l = M_l^{-1} \bar{q}_l + (k_{p_l} e_l + k_{i_l} \int e_l), \text{ and}$$

$$u_r = p_r = M_r^{-1} \bar{q}_r + (k_{p_r} e_r + k_{i_r} \int e_r).$$

4.2 Testing and Results

The sprayer is a closed loop system, delivering a desired flow rate using an error-based controller. The performance of the system is highly dependent on the accuracy of each flow sensor. System performance verification was performed concurrently with system performance testing. The testing occurred on a nearby farm in an environment that would likely resemble the rover and sprayer system's natural area of operation: one with external factors such as wind, dust, and uneven, rocky terrain. The testing occurred while the rover was in motion at a speed within the range of $0.2m/s$ to $0.4m/s$. In order to determine the performance of the system, time-response plots are captured for arbitrary step-inputs.

In Fig.8, the desired flow is set to 131 L/h for both banks. It takes the left and right bank approximately 8.5 seconds to settle

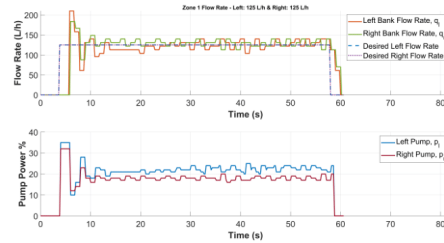


FIGURE 8: Top plot shows the left and right bank flow rate transient response to a step desired flow rate input of 131 l/h for both left and right bank. Bottom plot shows pump powers computed by the closed-loop controller.

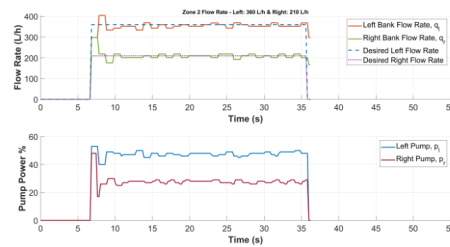


FIGURE 9: Top plot shows the left and right bank flow rate response to a step desired flow rate input of 360 l/h for the left bank and 210 l/h for the right bank. Bottom plot shows pump powers computed by the closed-loop controller.

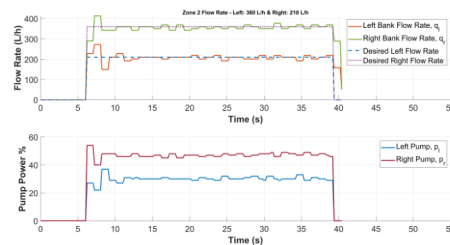


FIGURE 10: Top plot shows the left and right bank flow rate response to a step desired flow rate input of 210 l/h for the left bank and 360 l/h for the right bank. Bottom plot shows pump powers computed by the closed-loop controller.

given a 15% error band. The reason behind the larger error band is because the system never fully settles below a 15% error. This is likely due limitations in the sensors resolution and computation. The recorded flow rate data appears to take on discrete values. In the steady state region of Fig.8, the recorded data takes on one of the following values: 113 L/h, 122 L/h, 131 L/h, a 140 L/h.

TABLE 2: Zone-level flow rates (L/h)

Zone	Left-bank flow			Right-bank flow		
	Des.	Sen.	Act.	Des.	Sen.	Act.
1	131	125.3	133.4	131.0	130.8	131.7
2	360.0	353.4	356.0	210.0	211.0	211.8
3	210.0	208.0	217.5	360.0	353.4	352.3

This indicates that the resolution of the sensor is roughly 9L/h per bit. Given this finding, performing Root Mean Squared (RMS) calculations on the entirety of data provides greater insight over the steady state behavior. The RMS value for the left bank in Fig.8 is 125.3 L/h, and the RMS value for the right bank is 130.8 L/h. The RMS value is compared against the desired flow rate of 131 L/h for both banks; it equates to an error of 4.8% and 0.1% for the left and right bank, respectively.

The verification portion involved collecting the dispensed water via collection bins and noting the total time elapsed during the trial. Average flow rates for each bank were then calculated by dividing the total volume of water collected in the bins by the total time of the trial. Note that this method of verification is prone to human error in measurement and reaction time, resolution error in the analog measuring tools, evaporation due to wind, etc. From the verification test, the calculated flow rates are determined to be 133.4 L/h and 131.7 L/h for the left and right banks, respectively. This amounts to an error of 1.8% and 0.5% between the desired flow rates and averaged flow rates of the system.

Further performance testing and verification is performed on the controller for different desired flow rates as shown in Fig.9 and Fig. 10. Further system performance testing is carried out for different desired flow rates as shown in Fig.9 and Fig. 10.

In Fig.9, the settling time for the system is shorter, and the error band is smaller; for a 10% error band, the settling time is approximately 2.5 seconds for both banks. The RMS values are 353.4L/h and 211.0L/h for the left and right bank, respectively. It equates to errors of 1.8% and -0.4%. For the verification test performed in Fig. 9, the desired flow rates are 360L/h and 210L/h, respectively. The average flow rates are calculated to be 356.0L/h and 211.8L/h for the left and right bank respectively: yielding errors of 1.1% and 0.8%. Furthermore, in Fig.10, a similar result can be seen when the desired flow rates for the left and right banks are swapped with the other. The 10% settling time is also approximately 2.5 seconds for both banks. The RMS values are 208.0L/h and 353.4L/h for the left and right bank, respectively. This yields errors of 0.9% for the left bank and 1.8% for right bank.

In Fig. 10 the desired flow rates for the left and right bank are 210L/h and 360L/h, respectively. The average flow rates for the left and right bank were 217.5L/h and 352.3L/h, amounting to errors of 3.6% and 2.1%. The results from the system performance tests indicate that the system is able to administer the desired flow rate within a 4.8% RMS error margin. (i.e. RMS flow sensor reading compared to the desired flow rate). The system performance verification shows that the administered flow rate is accurate to within a 3.6% error margin (i.e. the physical amount of water collected over a finite time compared to the desired flow rate)

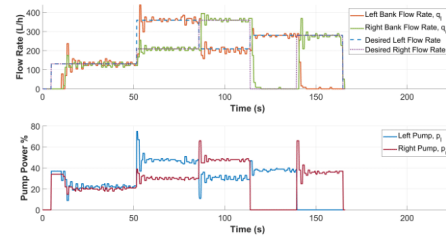


FIGURE 11: Multi-zone time response.

Figure 2 shows a performance test for a multi-zone trial. Five arbitrary zones were created, each with different desired flow rates. The multi-zone trial was performed while the rover was moving at speeds between 0.2m/s and 0.4m/s . The zones were advanced via button toggling through the main rover controller; this displays the potential ability to control the sprayer through an outer, rover control loop which could include GPS data, LIDAR, or vision data as shown in Fig. 3. The performance of the multi-zone run is consistent with what was seen in the step-input responses. The settling time between zone transitions is between 2 to 3 seconds. One notable exception can be seen in the plot during the initial power up; the reason behind the larger than expected settling time of the first zone is due to the pumps needing time to prime the system. Left at rest for a sufficient amount of time, the water drains downward and out of pumps, thus causing a larger initial settling time.

5. CONCLUSION

We have designed and constructed a modular sprayer attachment for the existing Agbot Rover which provides the necessary infrastructure for the mission. The closed-loop controller of the sprayer has been shown to be capable of administering the desired flow rate within an error margin between 0.1% and 4.8%. Lastly, a framework has been created between the sprayer and rover which will allow for the exchange of information between the two. The main purpose of the framework is to develop prescription-based controller.

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Figure A.7